

# Inference and linear combinations

## Last time: Activity

Coefficients:

|               | Estimate  | Std. Error | z value | Pr(> z ) |     |
|---------------|-----------|------------|---------|----------|-----|
| (Intercept)   | -3.989979 | 1.139951   | -3.500  | 0.000465 | *** |
| gre           | 0.002264  | 0.001094   | 2.070   | 0.038465 | *   |
| gpa           | 0.804038  | 0.331819   | 2.423   | 0.015388 | *   |
| factor(rank)2 | -0.675443 | 0.316490   | -2.134  | 0.032829 | *   |
| factor(rank)3 | -1.340204 | 0.345306   | -3.881  | 0.000104 | *** |
| factor(rank)4 | -1.551464 | 0.417832   | -3.713  | 0.000205 | *** |

Given your friend's GPA (3.95), school (rank 3), and current GRE score (590), what is your estimated probability that they will be accepted to graduate school?

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The standard deviation of GRE scores is 115.5 points. If your friend re-took the GRE and increased their score by one standard deviation, what would be the estimated change in their *odds* of being accepted?

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If your friend re-took the GRE and increased their score by one standard deviation, what would be their new estimated *probability* of getting accepted?

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Calculate a 95% confidence interval for the change in *odds* of acceptance associated with a *one-unit* increase in GRE score, holding GPA and school rank fixed

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Now calculate a 95% confidence interval for the change in *odds* of acceptance associated with a one *standard deviation* increase in GRE score, holding GPA and school rank fixed

## Activity

95% confidence interval for change in *odds* of acceptance associated with a one *standard deviation* increase in GRE score, holding GPA and school rank fixed:

$(1.014, 1.66)$

Your friend is reluctant to retake the exam unless they believe that their odds of acceptance will increase by at least 25%.

Would you recommend that your friend re-takes the exam?

## A new question

So far:

- + Inference for the *change* in log-odds ( $\beta_1$ ) or odds ( $e^{\beta_1}$ ) for a one-unit increase in  $X$
- + Inference for change in the log-odds and odds for a one *standard deviation* increase in  $X$

New question:

Can we do inference about a *probability*? E.g., can we construct a confidence interval for the true probability of admission, given values of the explanatory variables?

## Example

For simplicity, suppose we just look at GRE score for now:

|             | Estimate  | Std. Error | z value | Pr(> z ) |     |
|-------------|-----------|------------|---------|----------|-----|
| (Intercept) | -2.901344 | 0.606038   | -4.787  | 1.69e-06 | *** |
| gre         | 0.003582  | 0.000986   | 3.633   | 0.00028  | *** |

For a student with a GRE of 590, the *estimated* probability of admission is  $\hat{\pi}_i = 0.313$ .

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- + create a confidence interval for the true *log odds* of admission for this student
- + transform the interval into probabilities

## Confidence interval for the log odds

For a student with a GRE of 590, the *estimated* log odds is:

$$\widehat{\log \text{ odds}} = \hat{\beta}_0 + \hat{\beta}_1(590)$$

A confidence interval for the *true* log odds is:

$$\widehat{\log \text{ odds}} \pm z^* SE(\widehat{\log \text{ odds}})$$

## Variance of a sum of random variables

**Fact:** For two variables  $U, V$  and constants  $a, b$ :

$$Var(aU + bV) = a^2Var(U) + b^2Var(V) + 2abCov(U, V)$$

where  $Cov(U, V) = \text{Correlation}(U, V) \cdot SD(U) \cdot SD(V)$

## Finding the variance and covariance

In R, `vcov` can be used to extract variances and covariances for estimated coefficients:

```
m1 <- glm(admit ~ gre, family = binomial,  
          data = grad_app)  
vcov(m1)
```

```
##              (Intercept)              gre  
## (Intercept)  0.3672815851 -5.877568e-04  
## gre         -0.0005877568  9.722061e-07
```

## Confidence interval for the log odds

For a student with GRE 590,  $SE(\widehat{\log \text{ odds}}) = 0.11$

95% confidence interval for log odds:

$$\widehat{\log \text{ odds}} \pm 1.96 SE(\widehat{\log \text{ odds}})$$

## Confidence interval for the probability

95% confidence interval for the log odds, for a student with GRE 590:

$$(-1.004, -0.572)$$

How would I turn this into a 95% confidence interval for the true *probability* of admission to graduate school?

## Class activity

Work on the class activity (handout).